Reciprocal counter

Input frequency: fi
Reference frequency: fr

$$f_i=\frac{1}{T_i} \qquad f_r=\frac{1}{T_r} \qquad N_C=\frac{T_G}{T_r} \qquad T_G=N\cdot T_i$$

Number of counts: NC
Number of periods: N

$$N_C=N\frac{f_r}{f_i} \qquad f_i=f_r\frac{N}{N_C} \qquad T_G=N\cdot\frac{1}{f_i} \qquad f_G=\frac{f_i}{N}$$

$$N_C=N\frac{f_r}{f_i} \qquad N_C+1=N\frac{f_r}{f_i+df_i} \qquad 1+N\frac{f_r}{f_i}=N\frac{f_r}{f_i+df_i}$$

$$(f_i+df_i)\cdot(1+N\frac{f_r}{f_i})=N\cdot f_r \qquad f_i+N\cdot f_r+df_i+N\cdot df_i\cdot\frac{f_r}{f_i}=N\cdot f_r$$

$$f_i+df_i+N\cdot df_i\cdot\frac{f_r}{f_i}=0 \qquad df_i(1+N\cdot\frac{f_r}{f_i})=-f_i \qquad df_i=\frac{-f_i}{(1+N\cdot\frac{f_r}{f_i})} \qquad df_i=\frac{-f_i}{(1+N\cdot\frac{f_r}{f_i})}$$

$$df_i+N\cdot df_i\cdot\frac{f_r}{f_i}=-f_i \qquad df_i=\frac{-f_i^2}{N\cdot f_r+f_i}$$

Resolution:

$$|df_i|=\frac{f_i^2}{N\cdot f_r+f_i}$$ Unit: Hz

$$|df_{rel}|=100\,\%\cdot\frac{|df_i|}{f_i}=100\,\%\cdot\frac{f_i}{N\cdot f_r+f_i}$$ Unit: %